Chapter 4: Trigonometry and the Unit Circle

MULTIPLE CHOICE

1. $\frac{4}{3}\pi$ radians is equal to how many degrees?

A. 240°
B. 150°
C. 420°
D. 330°

ANS: A DIF: Easy OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: radians | degrees

2. The exact radian measure for an angle of 255° is

A. $\frac{17}{12}\pi$
B. $\frac{12}{17}\pi$
C. $\frac{17}{6}\pi$
D. $\frac{6}{17}\pi$

ANS: A DIF: Easy OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: radians | degrees

3. Which of the following angles, in degrees, is coterminal with, but not equal to, $\frac{6}{5}\pi$ radians?

A. 396°
B. 576°
C. 486°
D. 216°

ANS: B DIF: Average OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: radians | degrees

4. Determine the arc length of a circle with radius 5.5 cm if it is subtended by a central angle of $\frac{5}{2}\pi$ radians. Round your answer to one decimal place.

A. 1.4 cm
B. 43.2 cm
C. 4.4 cm
D. 6.9 cm

ANS: B DIF: Easy OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: radians | degrees

5. Determine the equation of a circle with centre at the origin and radius 8.

A. $x^2 + y^2 = 16$
B. $x^2 + y^2 = 64$
C. $x^2 + y^2 = 8$
D. $x^2 + y^2 = \sqrt{8}$

ANS: B DIF: Easy OBJ: Section 4.2 NAT: T2
TOP: Unit Circle KEY: unit circle | unit circle equation

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6. Determine the equation of a circle with centre at (3, –3) and radius 10.

A. $(x-3)^2+(y+3)^2=100$

B. $(x-3)^2+(y+3)^2=20$

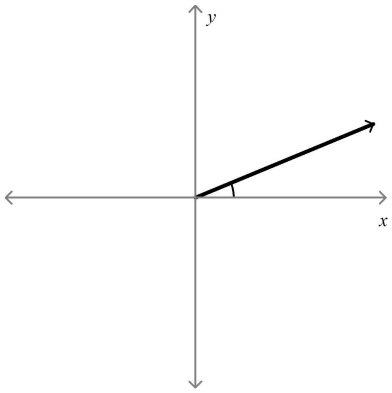
C. $(x-3)^2+(y+3)^2=10$

D. $(x-3)^2+(y+3)^2=\sqrt{10}$

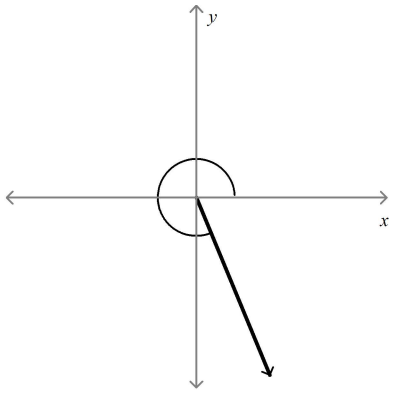
ANS: A DIF: Difficult + OBJ: Section 4.2 NAT: T2
TOP: Unit Circle KEY: unit circle | unit circle equation

7. Which graph represents an angle in standard position with a measure of $\frac{5}{8}\pi$ rad?

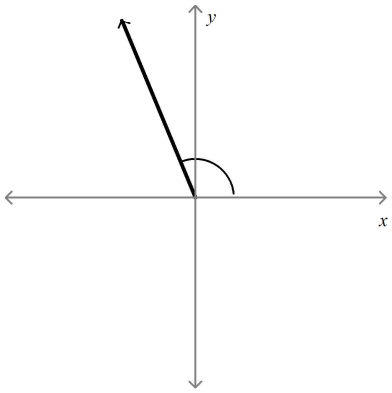
A.



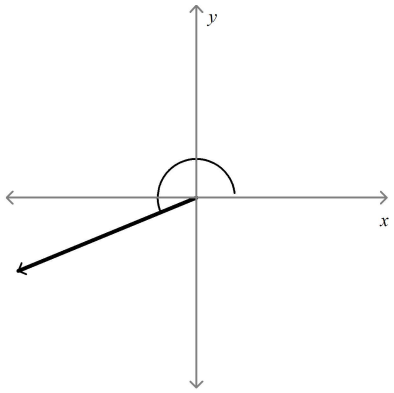
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B.



D.

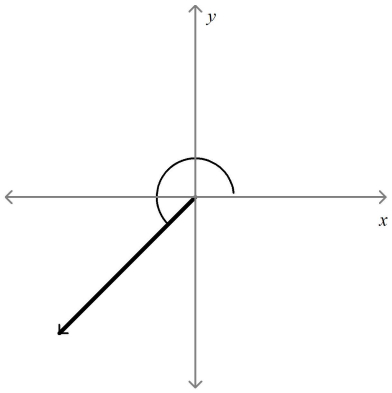


ANS: B DIF: Easy OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: radians

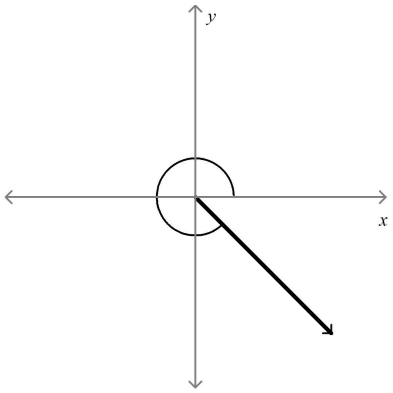
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8. Which graph represents an angle in standard position with a measure of 135°?

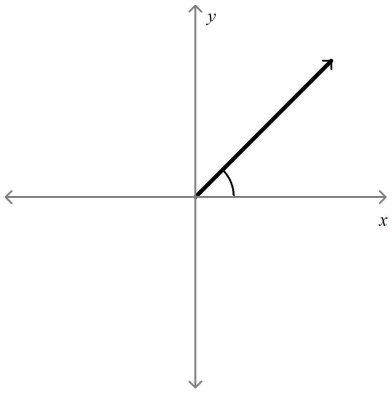
A.



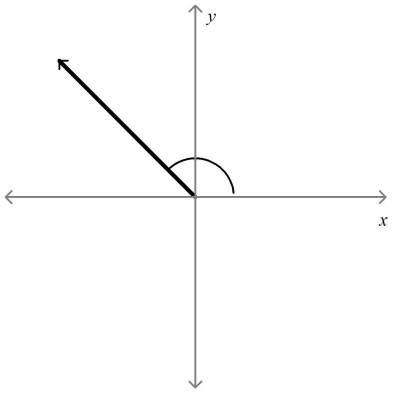
C.



B.



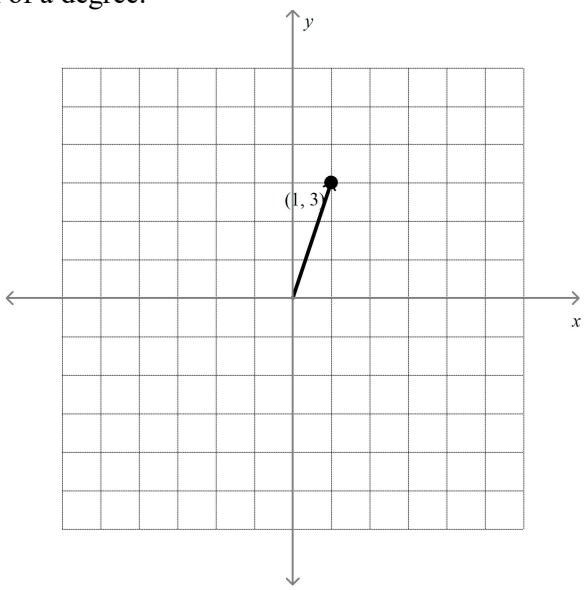
D.



ANS: D DIF: Easy OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: degrees

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9. Determine the measure of the angle in standard position shown on the graph below. Round your answer to the nearest tenth of a degree.



A. 161.6°
B. 341.6°
C. 71.6°
D. 251.6°

ANS: C DIF: Easy OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: radians

10. John cuts a slice from a circular ice cream cake with a diameter of 24 cm. His slice is in the shape of a sector with an arc length of 7 cm. What is the measure of the central angle of the slice, in radians? Round your answer to two decimal places, if necessary.

A. 1.71 rad
B. 3.43 rad
C. 0.29 rad
D. 0.58 rad

ANS: D DIF: Average OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: angle | radians | arc length

11. The coordinates of the point that lies at the intersection of the terminal arm and the unit circle at an angle of 110° are

A. (0.94, –0.34)
B. (–0.34, –2.75)
C. (–0.34, 0.94)
D. (–2.75, 0.94)

ANS: C DIF: Average OBJ: Section 4.3 NAT: T2
TOP: Trigonometric Ratios
KEY: trigonometric ratios | unit circle | terminal arm | angle

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12. Identify the point on the unit circle corresponding to an angle of 300° in standard position.

A. $(-\sqrt{3}, -\frac{\sqrt{3}}{2})$

B. $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$

C. $(\frac{1}{2}, -\frac{\sqrt{3}}{2})$

D. $(\frac{1}{2}, -\sqrt{3})$

ANS: C DIF: Average OBJ: Section 4.3 NAT: T2
TOP: Trigonometric Ratios KEY: exact value | unit circle
NOT: tan90 and tan 270 changed to remove undefined

13. Identify the point on the unit circle corresponding to an angle of $\frac{4\pi}{3}$ radians in standard position.

A. $(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$

B. $(\sqrt{3}, -\frac{\sqrt{3}}{2})$

C. $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$

D. $(-\frac{1}{2}, \sqrt{3})$

ANS: C DIF: Average OBJ: Section 4.3 NAT: T2
TOP: Trigonometric Ratios KEY: exact value | unit circle | radians
NOT: tan90 and tan270 do not include undefined

14. Which point on the unit circle corresponds to $\tan\theta = \sqrt{3}$?

A. $(-\frac{1}{2}, \sqrt{3})$

B. $(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$

C. $(\sqrt{3}, -\frac{\sqrt{3}}{2})$

D. $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$

ANS: D DIF: Average OBJ: Section 4.3 NAT: T2
TOP: Trigonometric Ratios KEY: Unit Circle | exact value | tangent ratio

15. Identify a measure for the central angle θ in the interval $0 \le \theta < 2\pi$ such that point $(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$ is on the terminal arm.

A. π

B. $\frac{\pi}{3}$

C. $\frac{3\pi}{4}$

D. $\frac{3\pi}{2}$

ANS: C DIF: Average OBJ: Section 4.3 NAT: T2
TOP: Trigonometric Ratios KEY: unit circle | exact value | tangent ratio

16. Which is a possible value of θ, to the nearest hundredth of a radian, when $\cos\theta = -0.58$?

A. –2.19

B. –0.62

C. 2.19

D. 0.84

ANS: C DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: reciprocal trigonometric ratios | approximate values

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17. If the angle θ is –5000° in standard position, it can be described as having made

A. $13\frac{8}{9}$ rotations

B. $-13\frac{8}{9}$ rotations

C. $27\frac{7}{9}$ rotations

D. $-27\frac{7}{9}$ rotations

ANS: B DIF: Average OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: rotations | standard position
NOT: Mixed numbers

18. During a routine, a figure skater completes $-1\frac{1}{9}$ rotations. How many degrees has the figure skater turned?

A. –400°

B. 400°

C. –220°

D. 580°

ANS: A DIF: Average OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: rotations | degrees | co-terminal angles
NOT: Mixed numbers

19. If the angle θ is 1600° in standard position, in which quadrant does it terminate?

A. quadrant III

B. quadrant IV

C. quadrant II

D. quadrant I

ANS: C DIF: Average OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: rotations | standard position

20. A ball is riding the waves at a beach. The ball's up and down motion with the waves can be described using the formula $h = 2.3\sin\left(\frac{\pi t}{3}\right)$, where h is the height, in metres, above the flat surface of the water and t is the time, in seconds. What is the height of the ball, to the nearest hundredth of a metre, after t = 17 s?

A. –0.87 m

B. –2.66 m

C. –1.99 m

D. 1.99 m

ANS: C DIF: Easy OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations KEY: trigonometric ratios

21. A tricycle has a front wheel that is 30 cm in diameter and two rear wheels that are each 12 cm in diameter. If the front wheel rotates through a angle of 32°, through how many degrees does each rear wheel rotate, to the nearest tenth of a degree?

A. 32.0°

B. 40.0ℑ

C. 80.0ℑ

D. 160.0ℑ

ANS: C DIF: Difficult + OBJ: Section 4.1 NAT: T1
TOP: Angles and Angle Measure KEY: arc length | degrees

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22. Determine the point in quadrant II where the line represented by $y = -3x$ intersects the unit circle.

A. (0.95, –0.32)
C. (–0.35, 0.94)
B. (–0.32, 0.95)
D. (–0.32, 0.94)

ANS: B DIF: Difficult + OBJ: Section 4.2 NAT: T2
TOP: Unit Circle KEY: unit circle | intersection point

23. The point P(0.391, 0.921) is the point of intersection of a unit circle and the terminal arm of an angle θ in standard position. What is the equation of the line passing through the centre of the circle and the point P? Round the slope to two decimal places.

A. $y = 2.36x$
C. $y = 2.36x + 0.92$
B. $y = 0.42x$
D. $y = 2.36x + 0.39$

ANS: A DIF: Difficult OBJ: Section 4.3 NAT: T3
TOP: Trigonometric Ratios KEY: unit circle | trigonometric ratios

24. Giai got an answer of 3.86 when she was calculating the value of a trigonometric function. Assuming Giai did her calculation correctly, which of the following was she calculating?

A. $\tan \frac{1}{12}\pi$
C. $\csc \frac{1}{12}\pi$
B. $\sec \frac{7}{12}\pi$
D. $\cot \frac{1}{12}\pi$

ANS: C DIF: Average OBJ: Section 4.3 NAT: T3
TOP: Trigonometric Ratios KEY: trigonometric ratios

25. A bottle is riding the waves at a beach. The bottle's up and down motion with the waves can be described using the formula $h = 1.4\sin\left(\frac{\pi t}{3}\right)$, where h is the height, in metres, above the flat water surface and t is the time, in seconds. When is the first time, to the nearest tenth of a second, that the height of the bottle will be 1.4 m?

A. 14.8 s
C. 0.9 s
B. 1.1 s
D. 1.5 s

ANS: D DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: trigonometric ratios | reciprocal trigonometric ratios

COMPLETION

1. The cosine of an angle is _____ in quadrant III.

ANS: negative

DIF: Easy OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: trigonometric ratios

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2. The sine of an angle is _____ in quadrant I.

ANS: positive

DIF: Easy OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: trigonometric ratios

3. The reciprocal of tangent is _____.

ANS: cotangent

DIF: Easy OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: trigonometric ratios | reciprocal trigonometric ratios

4. For a point on the unit circle, the y-coordinate can be determined using the _____.

ANS: sine of the reference angle

DIF: Easy OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: trigonometric ratios

5. The smallest non-negative value for sin x occurs when the angle in standard position is _____.

ANS: 180°

DIF: Easy OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: trigonometric ratios

MATCHING

Match each trigonometric expression with an equal expression to form a trigonometric identity.

A. $\sin\left(\frac{9\pi}{2} - \theta\right)$

B. $\cos\left(\frac{13\pi}{2} - \theta\right)$

C. $\cos\left(\theta + \frac{5\pi}{2}\right)$

D. $\tan\left(\theta + \frac{9\pi}{2}\right)$

E. $\tan\left(\frac{5\pi}{2} - \theta\right)$

F. $\sin\left(\theta + \frac{11\pi}{2}\right)$

1. $-\sin\theta$
2. $\cot\theta$
3. $\cos\theta$
4. $-\cot\theta$
5. $\sin\theta$

1. ANS: C DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: trigonometric ratios | translations | identity

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2. ANS: E DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: trigonometric ratios | translations | identity

3. ANS: A DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: trigonometric ratios | translations | identity

4. ANS: D DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: trigonometric ratios | translations | identity

5. ANS: B DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: trigonometric ratios | translations | identity

SHORT ANSWER

1. A child swings on a playground swing set. If the length of the swing's chain is 3 m and the child swings through an angle of $\frac{\pi}{9}$, what is the exact arc length through which the child travels?

ANS:

$a = \theta r$

$= \frac{\pi}{9}(3)$

$a = \frac{\pi}{3}$

The child travels through an arc length of $\frac{\pi}{3}$ m.

DIF: Easy OBJ: Section 4.1 NAT: T1 TOP: Angles and Angle Measure
KEY: arc length

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2. Determine the exact value of $\dfrac{\cot\left(\frac{\pi}{3}\right)+\csc\left(\frac{\pi}{3}\right)}{\cos\left(\frac{\pi}{4}\right)}$.

ANS:

$$\frac{\cot\left(\frac{\pi}{3}\right)+\csc\left(\frac{\pi}{3}\right)}{\cos\left(\frac{\pi}{4}\right)} = \frac{\frac{\cos\left(\frac{\pi}{3}\right)}{\sin\left(\frac{\pi}{3}\right)}+\frac{1}{\sin\left(\frac{\pi}{3}\right)}}{\cos\left(\frac{\pi}{4}\right)}$$

$$= \frac{\frac{\cos\left(\frac{\pi}{3}\right)+1}{\sin\left(\frac{\pi}{3}\right)}}{\cos\left(\frac{\pi}{4}\right)}$$

$$= \frac{\cos\left(\frac{\pi}{3}\right)+1}{\sin\left(\frac{\pi}{3}\right)\cos\left(\frac{\pi}{4}\right)}$$

$$= \frac{\frac{1}{2}+1}{\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)}$$

$$= \frac{\frac{3}{2}}{\frac{\sqrt{6}}{4}}$$

$$= \left(\frac{3}{2}\right)\left(\frac{4}{\sqrt{6}}\right)$$

$$= \frac{6}{\sqrt{6}}$$

$$\frac{\cot\left(\frac{\pi}{3}\right)+\csc\left(\frac{\pi}{3}\right)}{\cos\left(\frac{\pi}{4}\right)} = \sqrt{6}$$

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DIF: Average OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: trigonometric ratios | special angles

3. A 3-m ladder is leaning against a vertical wall such that the angle between the ground and the ladder is $\frac{\pi}{3}$. What is the exact height that the ladder reaches up the wall?

ANS:
Use the trigonometry of right triangles. The hypotenuse is the length of the ladder, or 3 m. The angle between the ladder and the ground is $\frac{\pi}{3}$. The opposite side to the angle is the height the ladder reaches up the wall. Let this height be *h*.

$$\frac{h}{3} = \sin\left(\frac{\pi}{3}\right)$$

$$h = 3\sin\left(\frac{\pi}{3}\right)$$

$$h = \frac{3\sqrt{3}}{2}$$

The height the ladder reaches up the wall is $\frac{3\sqrt{3}}{2}$ m.

DIF: Average OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: special angles | trigonometric ratios

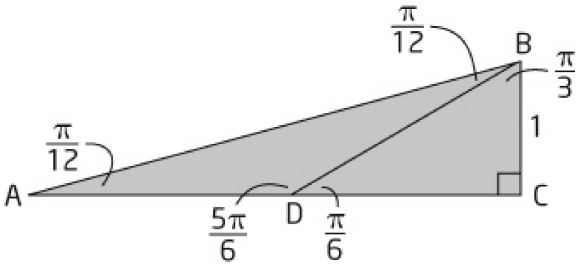
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4. Find the exact value of $\sin\left(\frac{11\pi}{12}\right)$.

ANS:

$$\sin\left(\frac{11\pi}{12}\right) = \sin\left(\pi - \frac{11\pi}{12}\right)$$

$$= \sin\left(\frac{\pi}{12}\right)$$



ΔBCD is a $\frac{\pi}{6}$-$\frac{\pi}{3}$-$\frac{\pi}{2}$ triangle, so BD = 2 and CD = $\sqrt{3}$. Since the measures of $\angle DAB$ and $\angle DBA$ are both $\frac{\pi}{12}$, ΔBDA is isosceles, and AD = BD = 2.

Find the measure of AB using the Pythagorean theorem.

$$AB^2 = AC^2 + BC^2$$

$$AB = \sqrt{AC^2 + BC^2}$$

$$= \sqrt{(2+\sqrt{3})^2 + 1^2}$$

$$= \sqrt{4 + 4\sqrt{3} + 3 + 1}$$

$$= \sqrt{8 + 4\sqrt{3}}$$

$$= 2\sqrt{2 + \sqrt{3}}$$

Thus,

$$\sin\left(\frac{\pi}{12}\right) = \frac{BC}{AB}$$

$$= \frac{1}{2\sqrt{2+\sqrt{3}}}$$

DIF: Difficult + OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: special angles | exact value | Pythagorean theorem

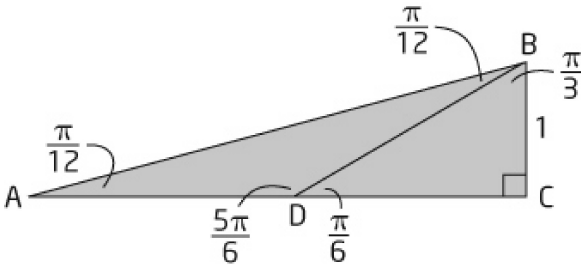
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5. Find the exact value of $\cos\left(\frac{5\pi}{12}\right)$.

ANS:

$$\cos\left(\frac{5\pi}{12}\right) = \sin\left(\frac{\pi}{2} - \frac{5\pi}{12}\right)$$

$$= \sin\left(\frac{\pi}{12}\right)$$



ΔBCD is a $\frac{\pi}{6}$-$\frac{\pi}{3}$-$\frac{\pi}{2}$ triangle, so BD = 2 and CD = $\sqrt{3}$. Since the measures of $\angle DAB$ and $\angle DBA$ are both $\frac{\pi}{12}$, ΔBDA is isosceles, and AD = BD = 2.

Find the measure of AB using the Pythagorean theorem.

$$AB^2 = AC^2 + BC^2$$

$$AB = \sqrt{AC^2 + BC^2}$$

$$= \sqrt{(2+\sqrt{3})^2 + 1^2}$$

$$= \sqrt{4 + 4\sqrt{3} + 3 + 1}$$

$$= \sqrt{8 + 4\sqrt{3}}$$

$$= 2\sqrt{2 + \sqrt{3}}$$

Thus,

$$\sin\left(\frac{\pi}{12}\right) = \frac{BC}{AB}$$

$$= \frac{1}{2\sqrt{2+\sqrt{3}}}$$

DIF: Difficult + OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: exact value | special angles | Pythagorean theorem

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6. Given that $\sin x = \cos\left(\frac{\pi}{5}\right)$ and that x lies in the first quadrant, determine the exact measure of angle x.

ANS:

$$\sin x = \cos\left(\frac{\pi}{5}\right)$$

$$\sin x = \cos\left(\frac{\pi}{2} - \frac{3\pi}{10}\right)$$

$$\sin x = \sin\left(\frac{3\pi}{10}\right)$$

$$x = \frac{3\pi}{10}$$

DIF: Average OBJ: Section 4.4 NAT: T4
TOP: Introduction to Trigonometric Equations
KEY: equivalent trigonometric expression | exact value

7. Without using a calculator, determine two angles between 0° and 360° that have a cosecant of $-\frac{2}{\sqrt{3}}$.

Include an explanation of how you determined the two angles.

ANS:

Since $\csc\theta = -\frac{2}{\sqrt{3}}$, $\sin\theta = -\frac{\sqrt{3}}{2}$. Since $\sin 60° = \frac{\sqrt{3}}{2}$, the reference angle is 60°. The ratio is negative in quadrants III and IV.This means that the angle can be found by looking for reflections of 60° that lie in these quadrants.
quadrant III: 180° + 60° = 240°
quadrant IV: 360° – 60° = 300°

DIF: Average OBJ: Section 4.3 NAT: T2 TOP: Trigonometric Ratios
KEY: reference angle | reciprocal trigonometric ratios | unit circle

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8. Given a circle of diameter 21 cm, determine the arc length subtended by a central angle of 1.2 radians.

ANS:

$$r = \frac{d}{2}$$

$$= \frac{21}{2}$$

$$= 10.5$$

$$a = r\theta$$

$$a = (10.5)(1.2)$$

$$= 12.6$$

The arc length is 12.6 cm.

DIF: Average OBJ: Section 4.1 NAT: T1 TOP: Angles and Angle Measure
KEY: arc length | central angle

9. Find the exact value of $\left[\cos\left(\frac{5\pi}{6}\right)\right]^2 - \left[\sin\left(\frac{5\pi}{6}\right)\right]^2$.

ANS:

$$\left[\cos\left(\frac{5\pi}{6}\right)\right]^2 - \left[\sin\left(\frac{5\pi}{6}\right)\right]^2 = \left(-\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$

$$= \frac{3}{4} - \frac{1}{4}$$

$$= \frac{1}{2}$$

DIF: Difficult OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: exact value | unit circle

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10. Angles A and B are located in the first quadrant. If $\sin A = \frac{\sqrt{2}}{2}$ and $\cos B = \frac{\sqrt{3}}{2}$, determine the exact value of sec A + sec B.

ANS:

$\sin A = \frac{\sqrt{2}}{2}$ $\quad\quad$ $\cos B = \frac{\sqrt{3}}{2}$

$\angle A = \frac{\pi}{4}$ $\quad\quad$ $\angle B = \frac{\pi}{6}$

$$\sec A + \sec B = \sec\left(\frac{\pi}{4}\right) + \sec\left(\frac{\pi}{6}\right)$$

$$= \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{3}}$$

$$= \frac{2\sqrt{3} + 2\sqrt{2}}{\sqrt{6}}$$

$$= \frac{2\sqrt{18} + 2\sqrt{12}}{6}$$

$$= \frac{2 \times 3\sqrt{2} + 2 \times 2\sqrt{3}}{6}$$

$$= \frac{3\sqrt{2} + 2\sqrt{3}}{3}$$

DIF: Average OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: exact value | reciprocal trigonometric ratios

11. Determine the exact measures for all angles where $\tan\theta = -\sqrt{3}$ in the domain $-180° \le \theta \le 180°$.

ANS:
The tangent ratio is negative in quadrants II and IV. In quadrant II for the domain $0° \le \theta \le 180°$, $\theta = 120°$. In quadrant IV for the domain $-180° \le \theta \le 0°$, $\theta = -60°$.

DIF: Difficult OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: primary trigonometric ratios | exact value

12. A grandfather clock shows a time of 7 o'clock. What is the exact radian measure of the angle between the hour hand and the minute hand?

ANS:
$\frac{5\pi}{6}$

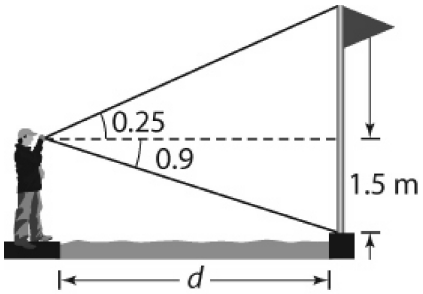
DIF: Easy OBJ: Section 4.1 NAT: T1 TOP: Angles and Angle Measure
KEY: radian

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PROBLEM

1. A 150-cm-tall person stands on the bank of a narrow river and observes a flagpole on the opposite bank. If the angle of elevation to the top of the flagpole is 0.25 and the angle of depression to the bottom of the flagpole is 0.09, determine the flagpole's height, in metres, to two decimal places.

ANS:
Let d represent the width of the river. Let h represent the height of the pole.



$$\tan 0.09 = \frac{1.5}{d} \qquad \tan 0.25 = \frac{h-1.5}{d}$$

$$d = \frac{1.5}{\tan 0.09} \qquad d = \frac{h-1.5}{\tan 0.25}$$

Equate and solve for h.

$$\frac{1.5}{\tan 0.09} = \frac{h-1.5}{\tan 0.25}$$

$$h = \frac{1.5\tan 0.25}{\tan 0.09} + 1.5$$

$$\approx 5.74$$

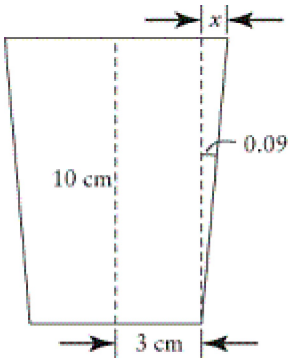
The height of the flagpole is approximately 5.74 m.

DIF: Difficult OBJ: Section 4.3 | Section 4.4 NAT: T3 |T5
TOP: Trigonometric Ratios | Introduction to Trigonometric Equations
KEY: radians | primary trigonometric ratios | trigonometric equations

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2. The volume of a drinking cup can be approximated by the formula $V = \frac{A_{\text{top}} + A_{\text{bottom}}}{2} h$, where the top and bottom of the cup are circular and h is the height. If the cup has a bottom diameter of 6 cm, a height of 10 cm, and sides that slope outward at an angle of 0.09, determine the cup's volume.

ANS:



First, find x.

$$\frac{x}{10} = \tan 0.09$$

$$x = 10\tan 0.09$$

$$A_{\text{bottom}} = \pi r^2 \qquad A_{\text{top}} = \pi r^2$$

$$= \pi(3)^2 \qquad = \pi(3+x)^2$$

$$= 9\pi \qquad = \pi(3+10\tan 0.09)^2$$

$$\approx 15.23\pi$$

$$V = \left(\frac{A_{\text{top}} + A_{\text{bottom}}}{2}\right) h$$

$$= \left(\frac{15.23\pi + 9\pi}{2}\right)(10)$$

$$\approx 381$$

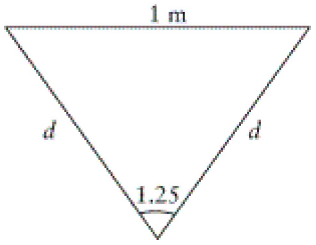
The volume of the cup is approximately 381 cm^3, or 381 mL.

DIF: Difficult OBJ: Section 4.3 | Section 4.4 NAT: T3 | T5
TOP: Trigonometric Ratios | Introduction to Trigonometric Equations
KEY: volume | trigonometric ratios | trigonometric equations | radians

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3. Two billiard balls collide and then separate from one another at the same, constant speed. Assume the billiard table is frictionless. The angle between the balls is 1.25 radians. After 2 s, the distance between the balls is 1 m. How fast are the balls moving, to the nearest hundredth of a metre per second?

ANS:



Use the cosine law.

$1^2 = d^2 + d^2 - 2d(d)\cos 1.25$

$1 = 2d^2 - 2d^2 \cos 1.25$

$= d^2(2 - 2\cos 1.25)$

$d = \sqrt{\dfrac{1}{2 - 2\cos 1.25}}$

Recall that $s = \dfrac{d}{t}$. The balls have travelled a distance, d, in metres, in a time, t, of 2 s.

$s = \dfrac{\sqrt{\dfrac{1}{2 - 2\cos 1.25}}}{2}$

≈ 0.43

Therefore, after 2 s, the billiard balls are moving at approximately 0.43 m/s.

DIF: Difficult + OBJ: Section 4.3 | Section 4.4 NAT: T3 | T5
TOP: Trigonometric Ratios | Introduction to Trigonometric Equations
KEY: trigonometric ratios | radians | cosine law

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4. To support a new 2.5-m wall in the construction of a home, the carpenters nail a piece of wood from the top of the wall to the floor, with the piece of wood forming the hypotenuse of the right triangle it makes with the wall and floor. The piece of wood is nailed to the ground such that it makes a 30° angle with the floor.

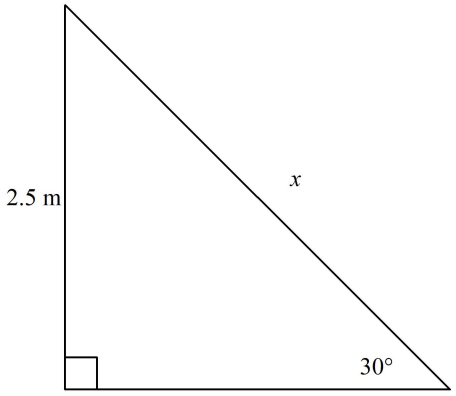
a) Represent this situation with a diagram.

b) Which trigonometric ratio can be used to determine the length of the piece of wood?

c) Determine the length of the piece of wood.

ANS:

a)



b) Use the sine ratio, since the side opposite the given angle is known and the hypotenuse is needed.

c) $\sin 30° = \frac{2.5}{x}$

$$x = \frac{2.5}{\sin 30°}$$

$$= \frac{2.5}{\frac{1}{2}}$$

$$= 5$$

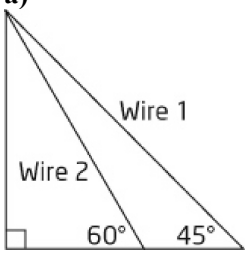
The length of the piece of wood is 5 m.

DIF: Easy OBJ: Section 4.3 | Section 4.4 NAT: T3 | T5
TOP: Trigonometric Ratios | Introduction to Trigonometric Equations
KEY: trigonometric ratios | special angles | trigonometric equations

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5. Two wires are connected to a tower at the same point on the tower. Wire #1 makes an angle of 45° with the ground and wire #2 makes an angle of 60° with the ground.
a) Represent this situation with a diagram.
b) Which wire is longer? Explain.
c) If the point where the two wires connect to the tower is 35 m above the ground, determine exact expressions for the lengths of the two wires.
d) Determine the length of each wire, to the nearest tenth of a metre.
e) How do your answers to parts b) and d) compare?

ANS:
a)



b) Both wires are connected to the tower at the same height, which is the opposite side to the given angles. Each wire length represents the hypotenuse of its respective triangle. The longer hypotenuse is the wire that forms the smaller angle, as it needs to be longer to reach the tower.

c) Wire 1: $\sin 45° = \frac{35}{x}$ Wire 2: $\sin 60° = \frac{35}{y}$

$$x = \frac{35}{\sin 45°} \qquad y = \frac{35}{\sin 60°}$$

$$= \frac{35}{\frac{1}{\sqrt{2}}} \qquad = \frac{35}{\frac{\sqrt{3}}{2}}$$

$$= 35\sqrt{2} \qquad = \frac{70}{\sqrt{3}}$$

$$= \frac{70\sqrt{3}}{3}$$

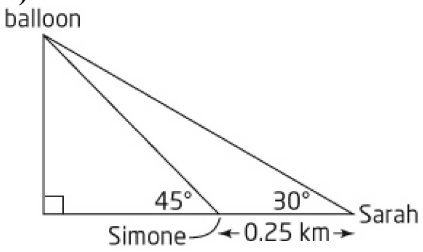
d) The length of wire #1 is 49.5 m and the length of wire #2 is 40.4 m.
e) The values calculated in part d) support the answers in part b).

DIF: Average OBJ: Section 4.3 | Section 4.4 NAT: T3 | T5
TOP: Trigonometric Ratios | Introduction to Trigonometric Equations
KEY: special angles | primary trigonometric ratios | right triangle

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6. Sarah and Simone are walking in a walk-a-thon down a straight street that leads to the finish line in the park. At the same time, they both notice a hot-air balloon directly over the finish line. Sarah sees the angle from the ground to the balloon as 30° and Simone, who is 0.25 km closer to the finish line than Sarah, sees the angle from the ground to the balloon as 45°.
a) Draw a diagram to represent this situation.
b) If x represents the distance that Simone is from the finish line, write an expression for the distance from Sarah to the finish line.
c) Write a trigonometric ratio for each girl's position that involves the height of the balloon, h; the distance, x, each girl is away from the finish line; and the angle from the girl to the balloon.
d) Rearrange each equation from part c) to isolate h.
e) Set the two expressions for h equal to each other and solve for x, to the nearest hundredth of a kilometre.
f) Determine the height of the balloon, to the nearest hundredth of a kilometre.

ANS:
a)



b) $x + 0.25$

c) Sarah: $\tan 30° = \dfrac{h}{x+0.25}$, Simone: $\tan 45° = \dfrac{h}{x}$

d) $h = (x+0.25)\tan 30°$ $\qquad$ $h = x\tan 45°$

$= \dfrac{x+0.25}{\sqrt{3}}$ $\qquad$ $= x$

e) $\dfrac{x+0.25}{\sqrt{3}} = x$

$x + 0.25 = \sqrt{3}x$

$\sqrt{3}x - x = 0.25$

$x(\sqrt{3} - 1) = 0.25$

$x = \dfrac{0.25}{\sqrt{3} - 1}$

≈ 0.34

f) $h = x$

≈ 0.34

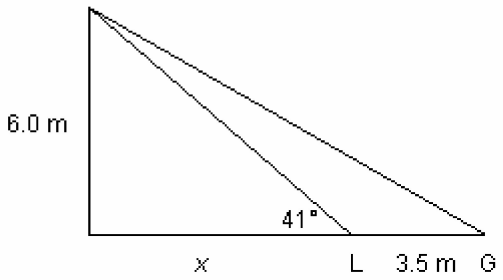
The height of the balloon is 0.34 km.

DIF: Average OBJ: Section 4.3 | Section 4.4 NAT: T3 | T5
TOP: Trigonometric Ratios | Introduction to Trigonometric Equations
KEY: two-dimensional problem | special angles | right triangle

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7. Gursant and Leo are both standing on the north side of a monument that is 6.0 m tall. Leo is standing 3.5 m closer to the monument than Gursant. Leo measures the angle from the ground to the top of the monument to be 41°. Determine the angle that Gursant would measure from the ground to the top of the monument, to the nearest degree.

ANS:
Draw a diagram of the situation.



Let x represent the distance that Leo is from the base of the monument.

$$\tan 41° = \frac{6.0}{x}$$

$$x = \frac{6.0}{\tan 41°}$$

Let $\angle A$ represent the angle that Gursant would measure. Let d represent the distance that Gursant is from the monument. Thus, $d = 3.5 + \frac{6}{\tan 41°}$.

$$\tan A = \frac{6}{3.5 + \frac{6}{\tan 41°}}$$

$$\angle A \approx 30°$$

Gursant would measure an angle of 30° from the ground to the top of the monument.

DIF: Difficult OBJ: Section 4.3 | Section 4.4 NAT: T3 | T5
TOP: Trigonometric Ratios | Introduction to Trigonometric Equations
KEY: two-dimensional problem | trigonometric ratios | right triangle

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8. The point (–5, 7) is located on the terminal arm of ∠A in standard position.
a) Determine the primary trigonometric ratios for ∠A.
b) Determine the primary trigonometric ratios for ∠B with the same sine as ∠A, but different signs for the other two primary trigonometric ratios.
c) Use a calculator to determine the measures of ∠A and ∠B, to the nearest degree.

ANS:
∠A is in quadrant II. Therefore, only the sine ratio will be positive.
Use the Pythagorean theorem.

$$r^2 = x^2 + y^2$$
$$= (-5)^2 + 7^2$$
$$= 25 + 49$$
$$= 74$$
$$r = \sqrt{74}$$

Therefore, $\sin A = \frac{7}{\sqrt{74}}$, $\cos A = -\frac{5}{\sqrt{74}}$, and $\tan A = -\frac{7}{5}$.

b) The quadrant in which the sine ratio is still positive, but the cosine and tangent ratios change from negative to positive, is quadrant I. In this quadrant, all three primary trigonometric ratios are positive.
$\sin B = \frac{7}{\sqrt{74}}$, $\cos B = \frac{5}{\sqrt{74}}$, and $\tan B = \frac{7}{5}$.

c) Use the fact that ∠B is the reference angle for ∠A.

$$\angle B = \sin^{-1}\frac{7}{\sqrt{74}}$$
$$\angle B \approx 54°$$
$$\angle A = 180° - 54°$$
$$= 126°$$

DIF: Average OBJ: Section 4.2 | Section 4.3 NAT: T2 | T3
TOP: Unit Circle | Trigonometric Ratios KEY: trigonometric ratios | reference angle | unit circle

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9. **a)** Without using a calculator, determine two angles between 0° and 360° that have a sine ratio of $-\frac{1}{2}$.

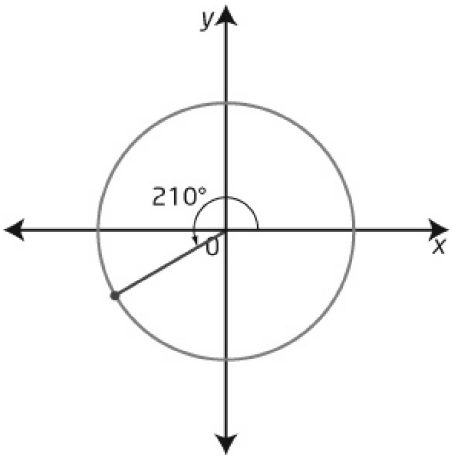
b) Use a calculator and a diagram to verify your answers to part a).

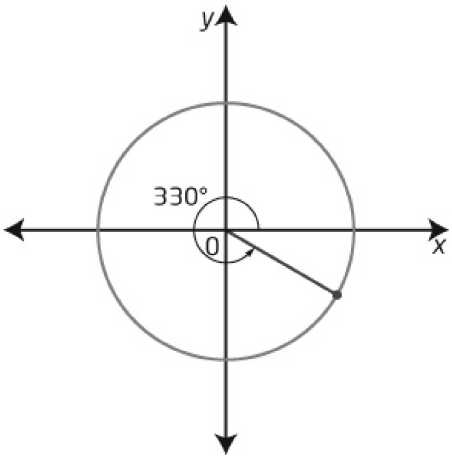
ANS:

a) Since sin 30° = $\frac{1}{2}$, the reference angle is 30°. The sine ratio is negative in quadrants III and IV.

Look for reflections of the 30° angle in these quadrants.
quadrant III: 180° + 30° = 210°
quadrant IV: 360° – 30° = 330°

b) Using a calculator, sin 210° = $-\frac{1}{2}$ and sin 330° = $-\frac{1}{2}$.





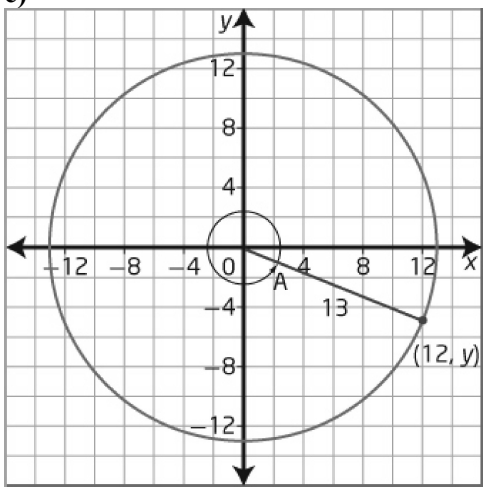
DIF: Average OBJ: Section 4.2 | Section 4.3 NAT: T2 | T3
TOP: Unit Circle | Trigonometric Ratios KEY: sine ratio | reference angle | unit circle

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10. Consider $\angle A$ such that $\cos A = \frac{12}{13}$.

a) In which quadrant(s) is this angle? Explain.
b) If the sine of the angle is negative, in which quadrant is the angle? Explain.
c) Sketch a diagram to represent the angle in standard position, given that the condition in part b) is true.
d) Find the coordinates of a point on the terminal arm of the angle.
e) Write exact expressions for the other two primary trigonometric ratios for the angle.

ANS:
a) Since the cosine ratio is positive, the angle is in quadrant I or IV.
b) If the sine ratio is negative, the angle is in quadrant IV.
c)



d) Use the Pythagorean theorem.

$r^2 = x^2 + y^2$

$13^2 = 12^2 + y^2$

$y^2 = 169 - 144$

$y^2 = 25$

$y = \pm 5$

Therefore, a point on the terminal arm is (12, –5).

e) $\sin A = -\frac{5}{13}$, $\tan A = -\frac{5}{12}$

DIF: Average OBJ: Section 4.2 | Section 4.3 NAT: T2 | T3
TOP: Unit Circle | Trigonometric Ratios KEY: trigonometric ratios | unit circle | reference angle

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11. In ΔABC, $c = 11$ cm, $b = 7$ cm, and $\angle B = 38°$.

a) Sketch possible diagrams for this situation.

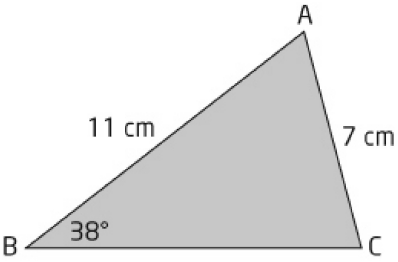
b) Determine the measure, to the nearest degree, of $\angle C$ in each diagram.

c) Find the measure of $\angle A$ in each diagram.

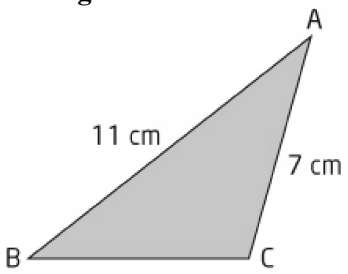
d) Calculate the length of BC in each diagram.

ANS:

a) **Triangle 1**



Triangle 2



b) For triangle 1,

$$\frac{\sin 38°}{7} = \frac{\sin C}{11}$$

$$\sin C = \frac{11 \sin 38°}{7}$$

$$\angle C \approx 75°$$

For triangle 2,

$$\angle C = 180° - 75°$$

$$= 105°$$

c) For triangle 1,

$$\angle A = 180° - (38° + 75°)$$

$$= 67°$$

For triangle 2,

$$\angle A = 180° - (38° + 105°)$$

$$= 37°$$

d) For triangle 1,

$$\frac{BC}{\sin 67°} = \frac{7}{\sin 38°}$$

$$BC = \frac{7 \sin 67°}{\sin 38°}$$

$$\approx 10.5$$

For triangle 2,

$$\frac{BC}{\sin 37°} = \frac{7}{\sin 38°}$$

$$BC = \frac{7 \sin 37°}{\sin 38°}$$

$$\approx 6.8$$

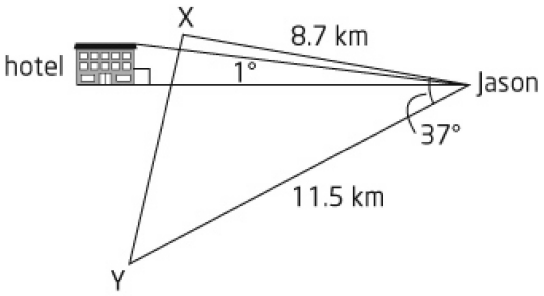
In triangle 1, the length of BC is 10.5 cm. In triangle 2, the length of BC is 6.8 cm.

DIF: Average OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: two-dimensional problem | ambiguous case | sine law

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12. Jason is standing 8.7 km from town X and 11.5 km from town Y. From where he stands, the angle between the two towns is 37°. A new hotel has just been built on the road connecting town X and town Y, exactly halfway between the two towns. From where Jason is standing, he sees that the angle of elevation to the top of the hotel is 1°. Determine the height of the hotel, to the nearest tenth of a metre. Include a diagram with your solution.

ANS:



First, use the cosine law to determine the distance between towns X and Y.

$(XY)^2 = 11.5^2 + 8.7^2 - 2(11.5)(8.7)\cos 37°$

$(XY)^2 \approx 48.133$

$XY \approx 6.94$

Then, use the cosine law to determine the measure of $\angle Y$.

$8.7^2 = 11.5^2 + 6.94^2 - 2(11.5)(6.94)\cos Y$

$75.69 = 180.4136 - 159.62\cos Y$

$$\cos Y = \frac{75.69 - 180.4136}{-159.62}$$

$\angle Y \approx 49.0°$

The hotel is built halfway between the two towns, so the distance from town Y to the hotel is 3.47 km. Next, use the cosine law to determine the distance from Jason to the hotel, JH.

$(JH)^2 = 11.5^2 + 3.47^2 - 2(11.5)(3.47)\cos 49°$

$(JH)^2 \approx 91.93$

$JH \approx 9.6$

Finally, determine the height of the hotel.

$$\tan 1° = \frac{h}{9.6}$$

$h = 9.6\tan 1°$

≈ 0.1676

The height of the hotel is approximately 0.1676 km, or 167.6 m.

DIF: Difficult + OBJ: Section 4.4 NAT: T5
TOP: Introduction to Trigonometric Equations
KEY: three-dimensional problem | cosine law | tangent ratio

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13. A rectangle has a diagonal of 8 cm. The diagonal creates a 60° angle at the base of the rectangle.
a) Write an exact expression for the base and the height of the rectangle.
b) Use your expressions to find the exact area of the rectangle.

ANS:

a) $\cos 60° = \frac{\text{base}}{8}$ and $\sin 60° = \frac{\text{height}}{8}$

$\text{base} = 8\cos 60°$ $\quad$ $\text{height} = 8\sin 60°$

$\text{base} = 8 \times \frac{1}{2}$ $\quad$ $\text{height} = 8\left(\frac{\sqrt{3}}{2}\right)$

$\text{base} = 4$ $\quad$ $\text{height} = 4\sqrt{3}$

b) $A = bh$

$= 4 \times 4\sqrt{3}$

$= 16\sqrt{3}$

The area of the rectangle is $16\sqrt{3}$ cm^2.

DIF: Easy OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: special angles | trigonometric ratios | area

14. Adriano knows that the distance from the point on the bridge where he is standing to one side of the canyon floor is 725 m, and that the distance from where he is to a point on the other side of the canyon floor is 816 m. If Adriano measures the angle between the two points on the canyon floor to be 11°, determine the width of the canyon, to the nearest tenth of a metre.

ANS:
Let x represent the width of the canyon. Use the cosine law.

$x^2 = 725^2 + 816^2 - 2(725)(816)\cos 11°$

$\approx 30\,019.72$

$x \approx 173.3$

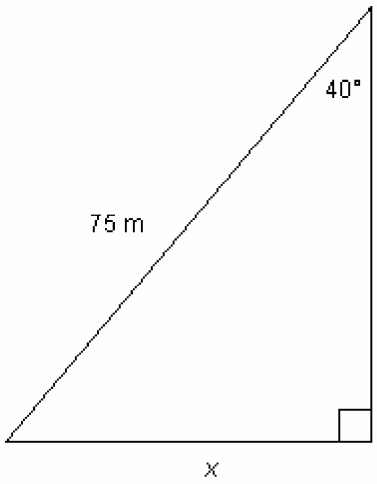
The width of the canyon is approximately 173.3 m.

DIF: Easy OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: cosine law | two-dimensional problem

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15. Simon is due north of a tall totem pole and walks 75 m, 40° west of south. From his new position, the totem pole is due east, with an angle of elevation of 17°. Determine the height of the totem pole, to the nearest tenth of a metre.

ANS:
Draw a diagram to find the distance, x, of Simon's new position from the totem pole.

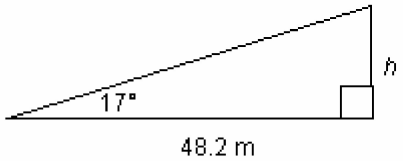


$$\sin 40° = \frac{x}{75}$$

$$x = 75 \sin 40°$$

$$\approx 48.2$$

Now, draw a diagram to show the angle of elevation of Simon's new position to the totem pole, and the height, h, of the totem pole.



$$\tan 17° = \frac{x}{48.2}$$

$$x = 48.2 \tan 17°$$

$$\approx 14.7$$

The height of the totem pole is approximately 14.7 m.

DIF: Average OBJ: Section 4.3 NAT: T3 TOP: Trigonometric Ratios
KEY: trigonometric ratios | three-dimensional problem

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16. **a)** For the given trigonometric ratio, determine two other angles that have the same value.
i) sin 45°
ii) tan 300°
iii) cos 120°
b) Explain how you determined the angles in part a).

ANS:
a) Answers may vary. Sample answers:
i) –315° and 405°
ii) –60° and 120°
iii) 240° and –240°
b) In all cases, locate the given angle on a Cartesian plane, and then identify the reference angle. Then, locate the other quadrant where the given trigonometric ratio has the same sign as the given ratio. Then, any angle that is co-terminal to the two angles in the diagram has the same trigonometric ratio as that given.

DIF: Average OBJ: Section 4.2 | Section 4.3 NAT: T2 | T3
TOP: Unit Circle | Trigonometric Ratios
KEY: trigonometric ratios | reference angle | co-terminal angles | unit circle

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17. The point P(–3, –6) lies on the terminal arm of an angle in standard position.
a) Which primary trigonometric ratios are positive and which are negative?
b) Which reciprocal trigonometric ratios are positive and which are negative?
c) Determine exact values for the primary trigonometric ratios.
d) Determine exact values for the reciprocal trigonometric ratios.

ANS:
a) Since the point on the terminal arm lies in quadrant III, the tangent ratio is positive, and the sine and cosine ratios are negative.
b) The cotangent ratio is positive, and the cosecant and secant ratios are negative.
c) $r^2 = x^2 + y^2$

$= (-3)^2 + (-6)^2$

$= 9 + 36$

$= 45$

$r = \sqrt{45}$

$= 3\sqrt{5}$

$\sin P = \frac{y}{r}$ $\cos P = \frac{x}{r}$ $\tan P = \frac{y}{x}$

$= \frac{-6}{3\sqrt{5}}$ $= \frac{-3}{3\sqrt{5}}$ $= \frac{-6}{-3}$

$= -\frac{2}{\sqrt{5}}$ $= -\frac{1}{\sqrt{5}}$ $= 2$

d) Using the answers in part c), take the reciprocal of each primary trigonometric ratio to write the reciprocal trigonometric ratios.

$\csc P = -\frac{\sqrt{5}}{2}$, $\sec P = -\sqrt{5}$, and $\cot P = \frac{1}{2}$

DIF: Average OBJ: Section 4.2 | Section 4.3 NAT: T2 | T3
TOP: Unit Circle | Trigonometric Ratios
KEY: primary trigonometric ratios | reciprocal trigonometric ratios | point on terminal arm

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18. Darren cuts a slice from his circular birthday cake, which has a diameter of 30 cm. The slice is in the shape of a sector with arc length 8 cm. What is the measure of the central angle of the slice, in radians, rounded to two decimal places?

ANS:
Since the diameter is 30 cm, the radius is 15 cm.

$a = r\theta$

$8 = 15\theta$

$\theta = \frac{8}{15}$

≈ 0.53

The central angle measures 0.53 rad.

DIF: Easy OBJ: Section 4.1 NAT: T1 TOP: Angles and Angle Measure
KEY: central angle | arc length

19. A bicycle tire revolves at 150 rpm (revolutions per minute). What is its angular velocity, in radians per second, rounded to two decimal places?

ANS:

$\frac{150 \text{ revolutions}}{\text{minute}} = \frac{150(2\pi)}{60 \text{ s}}$

$\approx 15.71 \text{ rad/s}$

The angular velocity is 15.71 rad/s.

DIF: Average OBJ: Section 4.1 NAT: T1 TOP: Angles and Angle Measure
KEY: angular velocity

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20. A group of marketing students are creating a strategy to sell ceiling fans. They determine that the number of sales will vary according to the month of the year. The formula $s = 9.2 + 2.4\cos\left[\frac{\pi}{6}(t-6)\right]$ gives the expected sales, in thousands, according to the month, t, where $t = 1$ represents January, $t = 2$ is February, and so on.

a) In which month are sales of fans expected to reach 11 600?

b) In which month are sales expected to be the least?

c) Does this formula seem reasonable? Explain.

ANS:

a) Solve the equation $11.6 = 9.2 + 2.4\cos\left[\frac{\pi}{6}(t-6)\right]$ for t.

$$11.6 = 9.2 + 2.4\cos\left[\frac{\pi}{6}(t-6)\right]$$

$$2.4 = 2.4\cos\left[\frac{\pi}{6}(t-6)\right]$$

$$1 = \cos\left[\frac{\pi}{6}(t-6)\right]$$

$$\frac{\pi}{6}(t-6) = 0 \text{ or } \frac{\pi}{6}(t-6) = 2\pi$$

$$t = 6 \text{ or } t = 18$$

These both represent June. Sales will be 11 600 in June. Since the cosine function is equal to 1 at this point, this is the maximum number of sales.

b) The minimum sales will occur when the cosine function is equal to −1, which occurs when

$$\frac{\pi}{6}(t-6) = \pi$$

$$t - 6 = 6$$

$$t = 12$$

The minimum sales are in December.

c) Yes. You would expect the most sales to occur at the beginning of summer and the least sales in a cold month like December.

DIF: Average OBJ: Section 4.4 NAT: T5
TOP: Introduction to Trigonometric Equations KEY: sine ratio | algebraic solution

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